

subsystem—propulsion, power, or payload—requires distinct security measures to safeguard against adversarial actions.

One infamous breach occurred in 2008 when the US Geological Survey's Landsat-7 satellite faced two hacking attempts. Hackers exploited ground station network vulnerabilities to gain unauthorised access. While intent remains unclear, the incident underscored the need to secure ground and space segments.

Another significant breach happened in 2017 when hackers attempted to control a commercial communications satellite. The thwarted attack revealed vulnerabilities in command systems, prompting industry-wide changes including:

- Implementation of zero-trust security models
- Enhanced authentication for command-and-control operations
- Development of AI-powered anomaly detection systems
- Regular firmware updates and security patches for satellite systems

Agencies have since learned valuable lessons. Encryption of uplink and downlink communications is now standard practice. Satellites are equipped with intrusion detection systems to monitor anomalies in command inputs. Ground stations employ zero-trust architectures, ensuring that even internal users are authenticated at every stage of interaction. Moreover, advancements in artificial intelligence have enabled predictive threat detection, allowing operators to respond proactively to potential breaches.

At the system level, satellites must incorporate multiple layers of security to protect against potential threats. This includes hardware-based security measures such as HSMs (hardware security modules), which securely store cryptographic

keys and perform essential cryptographic operations. By leveraging HSMs, satellite operators can mitigate risks associated with key management and ensure that sensitive data remains protected from unauthorised access.

Despite advancements, breaches persist. One incident involved hacking a commercial satellite operator's network, disrupting services for multiple clients. Investigations revealed inadequate protocols permitted undetected access. The breach led to stricter access controls and monitoring.

Lessons highlight the need for continuous monitoring and updating protocols. Organisations must maintain proactive cybersecurity postures, regularly assessing vulnerabilities and applying patches to pre-empt exploitation.

The human factor in satellite security

Technological advancements notwithstanding, the human factor remains a critical vulnerability. In 2022, a disgruntled employee at a satellite manufacturing firm attempted to sell proprietary encryption keys on the dark web. The breach was averted but highlighted the importance of insider threat monitoring and employee training.

Organisations have adopted stringent policies to address such risks. Background checks, continuous user behaviour monitoring, and regular cybersecurity awareness programmes are integral to satellite security. Combined with technical safeguards, these create comprehensive defences against internal and external threats.

Lessons from major security breaches

Satellite security history includes cautionary tales. The 2022 Viasat KA-SAT attack disrupted internet

services in Ukraine during a critical conflict period. Attackers exploited ground station network misconfigurations, causing widespread outages.

Viasat and stakeholders responded with advanced network segmentation and real-time monitoring tools, collaborating with government agencies and cybersecurity firms to address systemic weaknesses. The incident reinforced the importance of public-private partnerships in securing Satcom systems.

In 2015, a weather satellite network suffered disruption when hackers exploited ground station software vulnerabilities, reinforcing the need to secure all system components, not just the space segment.

In 2020, a telecommunications satellite exhibited unexpected behaviour due to a suspected cyberattack. Investigations traced the entry point to a compromised ground station, leading to new security protocols for ground segment operations.

Building a secure future for satellite communications

As reliance on satellite communications grows, so does the need for a robust security framework. Space agencies and private firms must adopt a multi-pronged approach, combining cutting-edge technology with strategic collaboration. Key measures include:

Enhanced encryption. Implementing quantum-resistant encryption algorithms to safeguard data against future threats

Resilient protocols. Regularly updating communication protocols like CCSDS to address emerging vulnerabilities

AI-driven security. Leveraging artificial intelligence to predict and counteract potential cyber threats

Global cooperation. Establishing